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11-year trends of psychological impact on Fukushima mothers and children post-nuclear accident

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ABSTRACT

This study examined changes in the mental health of mothers and children living in low-dose radiation areas over 5 years following the Fukushima Daiichi Nuclear Power Plant accident and 11 years after the accident. The mothers' and children's psychological symptoms, such as stress, depression, and anxiety about radiation, were particularly strong immediately after the accident and diminished with time. However, the survey conducted 11 years after the accident showed that those symptoms remained higher among mothers and children in Fukushima than in unaffected prefectures, indicating the impact persisted for at least 11 years. An age-period-cohort analysis of children's psychological symptoms showed a weak cohort effect, indicating that preschool children were greatly affected at the time of the accident. The results suggest that it is necessary to provide support aimed at improving the mental health not only of residents who evacuated from highly contaminated areas following the nuclear accident but also of residents in low-dose radiation-contaminated areas where evacuation orders were not issued. Moreover, the findings underscore the importance of monitoring the psychological development of children who grew up in these areas.

This study conducted a survey over a 5-year period following the accident at Tokyo Electric Power Company's Fukushima Daiichi Nuclear Power Plant (FDNPP). It examined the psychological effects of the nuclear accident on children aged 3–12 years and their mothers living in Fukushima City, which became an area contaminated with low-dose radiation after the accident. Additionally, we examined the situation 11 years after the accident.

The magnitude 9.0 earthquake that struck off the Pacific coast of the Tohoku region on March 11, 2011, and consequent tsunami caused meltdowns in multiple FDNPP reactors, which subsequently leaked large amounts of radioactive materials into the atmosphere. The materials that leaked on March 14–15 were deposited in various areas of Fukushima Prefecture, including Fukushima City, and residents were exposed to low doses over a wide area. According to a United Nations Scientific Committee on the Effects of Atomic Radiation (2022) report, the average estimated effective dose of radiation during one year immediately after the accident was 5.3 mSv for 1-year-old children, and 4.5 mSv for 10-year-old children in Fukushima City. The average effective dose in Fukushima City was not regarded as a level that would cause negative physical health effects on children or adults. However, low-dose exposure may have negative effects on the mental health of mothers and their preschool to school-age children, and these effects may persist over the medium to long term.

The Report of the UN Chernobyl Forum Expert Group "Health" (Bennett, Repacholi, & Carr, 2006) concluded that "the mental health impact of Chernobyl is the largest public health problem unleashed by

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the accident to date" (p. 95) based on the results of various studies on the health effects. The sufferers of the accident continued to show the high level of anxiety and medically unexplained physical symptoms for a long time, despite many efforts to provide accurate information by governments, scientists, and international organizations (Bennett et al., 2006; International Atomic Energy Agency, 2006).

Furthermore, the negative impacts on mental health that appear to be attributable to nuclear accidents are not limited to evacuees who lived near nuclear power plants or in highly contaminated areas. After the Three Mile Island (TMI) nuclear power plant accident, Dohrenwend et al. (1981) interviewed people living within a 20-mile (32.19 km) radius of the nuclear power plant to assess their level of distress. The results indicated that among patients with chronic mental disorders in community mental health centers who had above-average levels of distress, 22 % were residents within 5 miles (8.05 km) of an evacuation advisory, and 15 % were residents outside the area. Lee (1996) summarized the findings of the 1990 International Chernobyl Project on stress symptoms, attitudes toward self, and beliefs about the future of residents of contaminated areas and residents of adjacent demographically similar clean control areas. Lee (1996) found that although stress and anxiety were statistically significantly higher in those living in contaminated areas than in the control group, absolute levels of stress and anxiety were high, and morale was low in both areas. Havenaar et al. (1996) examined the relationship between the mental health of residents of Gomel region, Belarus, which is approximately 100 km northeast of Cholnobyl nuclear power plant (ChNPP), and contamination levels 6 years after the nuclear disaster. The results showed that 62.7 % of the residents living in low-contamination areas (0–185 KBq Cs^{137}/m^2) outside the radiation control areas had mental health problems or some types of symptoms (GHQ-12 score of 2 or higher). The prevalence of these disorders in the areas was not much lower than 67.8 % in the more contaminated Zones 2 (185–555 KBq Cs^{137}/m^2) and 71.6 % in Zone 3 (>555KBq Cs^{137}/m^2). Danzer and Danzer (2016) investigated the subjective well-being and mental health of people living in low-contamination areas (with an average exposure dose from May to December 1986 of 1 mSv) 20 years after the ChNPP accident and found that the respondents had low subjective well-being and depressive symptoms 20 years after the catastrophe. Additionally, the negative impacts on mental health that appear to be attributable to nuclear accidents are long-lasting, even for low-dose exposures, as shown in previous studies (Danzer & Danzer, 2016; Havenaar et al., 1996).

Moreover, gender differences have been observed in the negative impact of nuclear accidents on mental health. Women, particularly mothers, were repeatedly reported to be more affected by radiation disasters than men in studies following the TMI and ChNPP accidents. This was also true for low-dose exposures. For instance, Dohrenwend et al. (1981) found that shadow evacuation was more common among women than men during the TMI accident, with 72.4 % of mothers with preschool-age children (51.8 % in the total sample) evacuating. Havenaar et al. (1996) found that being a mother of a child under 18 years was a risk factor for nuclear-related anxiety disorders.

However, the effects of nuclear accidents on children are poorly understood. Much of the research on the effects on children has focused on the effects of exposure during the fetal stage and infancy from the ChNPP accident on children's cognitive development and mental health problems; studies have examined children who experienced accidents during fetal life or infancy at 6–8 years of age (Igumnov & Drozdovitch, 2000; Kolominsky, Igumnov, & Drozdovitch, 1999; Nyagu, Loganovsky, & Loganovskaja, 1998) or after 10 years of age (Bromet et al., 2011; Bromet, Gluzman, Schwartz, & Goldgaber, 2002; Heiervang, Mednick, Sundet, & Rund, 2011; Huizink et al., 2007; Litcher et al., 2000; Taormina et al., 2008), with inconsistent results. Researchers from Belarus and Ukraine conducted follow-up studies 6–12 years later, on children who experienced the ChNPP accident in utero and found that the well-being of the group exposed to radiation during the prenatal period was poorer than that of the unexposed control group (Igumnov & Drozdovitch, 2000; Kolominsky et al., 1999; Nyagu et al., 1998). Huizink et al. (2007) investigated 14-year-old children from Finland[2] who were exposed in utero and found that the exposed group had more behavioral problems than the control group. Furthermore, some studies have examined the well-being and psychopathology of children who were fetal age to 15 months old at the time of the accident and their mothers, and while these studies indicated that the mothers were psychologically affected 11 and 19 years after the accident (Adams et al., 2002; Adams, Guey, Gluzman, & Bromet, 2011; Bromet et al., 2002), there are also findings suggesting that the children were not psychologically affected at any time (Bromet et al., 2000; Bromet et al., 2011; Litcher et al., 2000). Although discrepancies exist in these results, the hypothesis that radiation disasters have medium- to long-term effects on children's mental health should not be dismissed. This is evidenced by the accumulated psychological findings on disasters and development, which demonstrate that children's reactions to disasters and resulting behaviors are mediated and adjusted by their relationship processes with their parents (Cobham, McDermott, Haslam, & Sanders, 2016; Gewirtz, Forgatch, & Wieling, 2008; Masten & Osofsky, 2010; Silverman & La Greca, 2002). Children are highly dependent on their parents and family, not only in infancy but also at school age and early adolescence. If the parental and family environments are disrupted by a disaster, this has a negative impact on children's development and mental health. Additionally, maternal anxiety, stress, and depression have been shown to inhibit positive, effective parenting practices and discipline for children (Chung, Lanier, & Wong, 2022; Epkins & Harper, 2016; Jackson & Choi, 2018; Liu & Wang, 2015).

Various studies conducted after the FDNPP accident partially support the hypothesis that radiation disasters, even at low-dose exposures, can have long-term impacts on children's well-being. According to a survey conducted by the Japanese Ministry of Education, Culture, Sports, Science and Technology (2013) on the effects of the disaster on children ranging from kindergarteners to high school students, the mental health of children in Fukushima was found to be more severely affected compared to children in Iwate and Miyagi Prefectures, who were also impacted by the earthquake and tsunami. The Fukushima Medical University has been conducting a continuous Mental Health and Lifestyle Survey (MHLS) since the accident, tracking the mental health of evacuated infants and children from evacuation zones using the Strength and Difficulties Questionnaire (SDQ). In 2011, the year of the accident, the proportion of children with behavioral and emotional problems was significantly higher than the standard values (Kamiya, Ohto, & Yasumura, 2021; Maeda et al., 2022). Although the proportion of children with behavioral and emotional problems decreased after the year following the accident, it took about five years to reach levels comparable to the standard data. The MHLS survey treats the mental health of parents as that of adults, and therefore, no specific data on parental mental health is provided. However, Maeda et al. (2022) suggest that the children's issues may be related to the worries of their mothers about their children.

While the MHLS data focuses on evacuees, Tsutsui, Ujiie, Takaya, and Tominaga (2020) and Sung (2015) conducted studies targeting children and their parents living in low-dose radiation-contaminated areas where evacuation orders were not issued. Tsutsui et al. (2020),

[2] Apart from the former Soviet Bloc, the Scandinavian countries were among the regions most severely affected by the Chernobyl accident. However, all areas had low-dose radiation contamination, with no concern regarding physical or genetic effects. Henriksen and Saxebøl (1988) estimated the additional annual exposure for the public in Norway to be 0.3 mSv. According to Heiervang et al. (2011), the external exposure dose for residents of highly contaminated areas during the 18 months following the accident was 0.935 mSv. Auvinen et al. (2001) estimated the additional exposure of the entire Finnish population in the 2 years following the accident to be 0.4 mSv. In the area with the highest level, their estimate 2 years after the accident was 1.0 mSv.

which began immediately after the FDNPP accident and continued for 5 years, examined trends in psychological effects in 18- and 42-month-old children and their mothers who lived in low-dose radiation-contaminated areas in Fukushima Prefecture. The results revealed that psychological symptoms, such as stress, depression, and post-traumatic stress disorder (PTSD), were present in mothers and children immediately after the accident and decreased over time; however, even in a survey conducted 5 years after the accident, significantly higher psychological effects were found in mothers and children who continued to live in low-dose radiation-contaminated areas of Fukushima than in areas unaffected by the accident. The results also showed that the mothers' anxiety about radiation persisted for 5 years or more after the accident. Tsutsui et al. (2020) argued that the psychological effects observed in children after the nuclear accident may be due to the behavior of their parents. Sung initiated a study in 2013 targeting children born in 2008 and their parents living in Nakadori, a region with low-dose radiation contamination (Sung, 2015). They assessed the children's mental health using the SDQ, similar to the MHLS. The results from the 2013 survey, conducted two years after the accident, indicated that the children had higher support needs across all five domains measured by the SDQ, compared to pre-accident surveys (Iwasaka, Matsuura, Maeda, & Nezu, 2010; Matsuishi et al., 2008), with particularly high support needs in the conduct domain. They also found that maternal depression and discrepancies between spouses regarding coping with radiation exposure significantly predicted children's conduct scores. In their subsequent follow-up study, Sung et al. (2021) discovered that the proportion of children requiring support, particularly boys, had plateaued and remained high even nine years after the accident, with fifth-grade students showing a higher proportion of children needing support compared to the national survey (Moriwaki & Kamio, 2014).

The findings of Tsutsui et al. (2020) and Sung (2015); Sung et al. (2021) suggest that the nuclear accident has had long-term effects on the mental health of not only evacuated infants and children but also those living in low-dose radiation-contaminated areas. Additionally, both Sung (2015) and Tsutsui et al. (2020) indicate, similar to the suggestions made by Maeda et al. (2022) for evacuated children, that the mental health issues observed in children from low-dose radiation-contaminated areas may also manifest through the mediation of parental mental health and behaviors. If children's mental health problems manifest through the mediation of parental mental health and behaviors, it is reasonable to assume that children born in low-dose radiation-contaminated areas after the disaster might also be affected in some way.

This study aims to investigate and analyze the psychological impact of the nuclear accident on children and parents living in low-dose radiation-contaminated areas from a developmental perspective. Developmental research on disasters has shown that the effects of disasters on children vary according to their age (Masten, Narayan, Silverman, & Osofsky, 2015; Masten & Osofsky, 2010). The psychological effects of the nuclear accident on the mental health of children and parents are likely to differ depending on the age of the child at the time of the accident, and they are also expected to be influenced by the child's developmental stage at the time of the survey and the time elapsed since the accident. However, as shown below, existing findings from the TMI, ChNPP, and FDNPP accidents have not sufficiently clarified the relationship between the psychological impact of the accident and the developmental stages of children.

Most of the studies on the effects of nuclear accidents have been conducted with pregnant or nursing mothers at the time of the accident, whereas data on the reactions of mothers of infants and school-aged children to the accidents and on changes over time are lacking. Shortly after the Three Mile Island nuclear accident, Dohrenwend et al. (1981) investigated the psychological reactions of children in grades 7, 9, and 11 within 10 miles of the TMI nuclear power plant. However, they did not examine the subsequent progression of these psychological responses. Several studies conducted after the ChNPP accident included individuals who were children at the time of the incident. For example, the study by Havenaar et al. (1996), conducted six years after the accident, included individuals who were 9 years old at the time of the accident. The study by Cwikel, Abdelgani, Goldsmith, Quastel, and Yevelson (1997) on individuals who had relocated to Israel from near the plant included those who were 10 years old at the time of the accident. Danzer and Danzer's (2016) samples, conducted 20 years after the accident, included individuals who were 0–18 years old at the time of the accident. However, these studies did not analyze the effect of the child's age at the time of the accident. Furthermore, the effects of the time elapsed since the accident and the age at the time of the survey were not analyzed. Developmental research indicates that the impact of disasters varies by children's age (Franks, 2011a; Masten et al., 2015; Narayan & Masten, 2012; Peek, Abramson, Cox, Fothergill, & Tobin, 2018; Silverman & La Greca, 2002). Thus, this effect may vary by age at the time of the accident, and the negative impact of accidents on mental health may change over time.

Moreover, although several longitudinal and follow-up studies have been conducted on the effects of the TMI and the ChNPP accidents on mothers and children, most of the follow-up studies were conducted nearly 10 years after the accidents or even later. However, data on the reactions of mothers and children to the accident and psychological effects of the accident during the 10-year period immediately following the accident are extremely limited. Dohrenwend et al. (1981) surveyed mothers of preschool children in the year of the TMI accident but did not conduct a follow-up study. Longitudinal studies have been conducted 9, 12, 30, and 42 months after the TMI accident (Bromet, Parkinson, & Dunn, 1990), and 10 years after it (Dew & Bromet, 1993). However, they do not report data on children. Furthermore, many of the follow-up studies on the mental health of children affected by the ChNPP accident during the fetal or infant stages were conducted years after the incident, with no information available from the time immediately following the accident to the time of the survey. Follow-up studies conducted by researchers in Ukraine and Belarus were carried out 4 to 12 years after the accident (Igumnov & Drozdovitch, 2000; Kolominsky et al., 1999; Nyagu et al., 1998). The study by Huizink et al. (2007) in Finland was conducted 14 years after the accident, while Bromet and collaborators conducted their studies 11 and 19 years after the accident (Adams et al., 2002; Adams et al., 2011; Bromet et al., 2002; Bromet et al., 2011; Litcher et al., 2000). Additionally, the study by Heiervang et al. (2011) in Norway was conducted 18 years after the accident.

Investigations into the effects of the FDNPP accident were initiated and conducted soon after the incident, allowing for near real-time tracking of the impact. Tsutsui et al. (2020) conducted a study over a five-year period following the nuclear accident, examining the psychological symptoms believed to be related to the incident among children aged 18 months and 42 months, as well as their mothers, at each assessment point. However, the age of the children in their study was limited to infants and toddlers, excluding information on children of other age groups. Sung (2015) and Sung et al. (2021) tracked children who were three years old at the time of the accident and their parents for a period of 10 years, but again, the study did not include data on children outside this specific age cohort. The HMLS followed children aged 4 to 15, but the focus was on evacuee children, and thus does not directly provide insights into the psychological effects of the accident on children and parents in areas with low-dose radiation contamination.

To examine the psychological impact of a nuclear accident on children and parents from a developmental perspective, it is important to widen the age range of the children at the time of the accident and to extend the follow-up period after the accident. By incorporating these two axes into the analysis, it becomes possible to understand how the psychological effects of the nuclear accident are influenced by the child's age at the time of the accident, the developmental stage of the child at the time of the survey, and the number of years elapsed since the accident.

Therefore, this study conducted the following two surveys: (1) For five years following the year of the accident, a yearly sampling of

children aged 3–12 and their mothers was conducted to assess both children's and mothers' mental health, and the mothers' anxiety about radiation. (2) In the eleventh year after the accident, a similar survey was conducted on children and their mothers. Through these surveys, the study analyzed the trends in the children's and mothers' mental health and psychological responses to the nuclear accident, based on the children's age (*Age*), the number of years elapsed since the accident (*Period*), and the children's birth year (*Cohort*). We explored whether the reaction to the event and its course differed depending on the age of the child and whether the reaction to the event and its course differed depending on the age of the child during the disaster. Based on the time-series changes identified in previous research, we performed an age-period-cohort (APC) analysis to identify changes in mental effects over time following the disaster.

Longitudinal research is recommended as a methodology when examining the impact of an event after a certain period or how it changes over time. However, longitudinal studies have some disadvantages. First, loss of participants is inevitable in longitudinal studies. When surveys are repeated and the follow-up period is extended, the attrition rate increases, and the sample size that includes data from all survey points tends to decrease. Second, the victims of a disaster cope with or adapt to the effects of the disaster over time. However, according to several studies conducted after the TMI accident, the effects, which diminish over time, increase again with major venting (Gatchel, Schaeffer, & Baum, 1985), restart of the remaining reactors (Dew, Bromet, Schulberg, Dunn, & Parkinson, 1987), and the accident's 10th anniversary (Dew & Bromet, 1993). Repeated requests for investigation to the same individuals may hinder the victims' ability to cope with and adapt to the effects of the accident. Moreover, the possibility that the survey records a "temporal" reaction due to recalling the accident rather than a "normal" psychological state at the time of the survey cannot be excluded. Thus, this study used a repeated cross-sectional survey to avoid the second problem.

Methods

Psychometric scale

To investigate the psychological effects of living in low-dose radiation-contaminated areas after the nuclear accident on mothers with small children, the Psychological Symptom Scale for Radiation Disaster for Mothers (Tsutsui et al., 2020) and Radiation Protection Behavior Rating Scale (Tsutsui et al., 2020) were applied. The former consists of eight questions related to PTSD, depressive symptoms, and psychological stress in the past month (Panel A in Table 1). Responses were rated on a four-point Likert scale (0 = *never*; 3 = *frequently*). Tsutsui et al. (2020) clarified the validity and reliability of the scale and reported that the scale has a one-factor structure. Therefore, the mean of the eight items was used to evaluate mothers' psychological symptom scores in this study.

The six-item Radiation Protection Behavior Rating Scale was used to assess the behaviors of mothers with children to avoid health hazards from radiation following the FDNPP accident and the intensity of their anxiety about radiation (Panel B in Table 1). Responses were rated on a three-point Likert scale (0 = *always*; 2 = *never*), with Q4 and Q6 reverse-scored. The mean score of the six items was calculated as the radiation protection behavior scores.

The Psychological Symptom Scale for Radiation Disaster for Children (Tsutsui et al., 2020) was used to assess the mental reactions of children living in areas contaminated with low-dose radiation. As the scale was designed for children up to 42 months of age, the questions were phrased carefully to avoid changing the contents and assess the behavior of children from 3 to 12 years of age (Table 2). The factor structure and reliability of the modified scale were examined in a preliminary study.

Table 1
Questionnaires used to examine mothers' psychological symptoms (adopted from Tsutsui et al., 2020).

Panel A: Psychological symptom scale for radiation disaster (for mothers)	
Q1.	I get irritated or angry easily.
Q2.	I get startled by sounds.
Q3.	It is difficult for me to focus on the work that I do on a regular basis.
Q4.	I suddenly recall things about the disaster (the Great East Japan Earthquake).
Q5.	I have trouble falling asleep or wake up in the middle of the night.
Q6.	I get depressed.
Q7.	I do not have an appetite, or I cannot control my appetite.
Q8.	I get tired easily and feel sluggish.

Panel B: Radiation protection behavior rating scale	
Q1.	How often do you put clothes outside to dry?
Q2.	How often do you use an air ventilator?
Q3.	How often do you open windows to circulate the air?
Q4.	How often are you careful about what the children drink?
Q5.	How often do you let the children play outside and go for walks?
Q6.	How often do you ensure the children wear a facemask when going out?

Preliminary research on the psychological symptom scale for radiation disaster for children

A preliminary survey was conducted approximately 3 months after the Great East Japan Earthquake on elementary schools and kindergartens in Koriyama City, Fukushima Prefecture. Koriyama is located approximately 50 km west of the FDNPP. It is approximately 40 km south of Fukushima City, where this survey was conducted, but is in the same Nakadori area as Fukushima City and is one of the areas that experienced low-dose radiation contamination. In cooperation with the Koriyama City Board of Education, a modified version of the Psychological Symptom Scale for Radiation Disaster for Children was distributed to parents with children aged 3–12 years, and 672 responses were received.

Data from 568 respondents were included in the analysis of the preliminary survey. Data with no gender reported and responses provided by individuals other than mothers were excluded from the analysis. An exploratory factor analysis using the principal factor method and Promax rotation was conducted to examine the factor structure of the scale. The analysis revealed that the scale consisted of three factors (Table 2). The structure was the same as that of the Psychological Symptom Scale for Radiation Disaster (for Children), which was created for children aged 18–42 months. Thus, the same factor names (Depressiveness, Anxiety and Fear, Anger and Restlessness) were used as in the scale developed for 18- to 42-month-old children. Table 2 shows the Cronbach's alpha coefficients of each factor. This scale was named the Psychological Symptom Scale for Radiation Disaster for Children aged 4–12 years; the mean score of each subscale was used in the analysis as the Depressiveness, Anxiety and Fear, and Anger and Restlessness scores.

Main survey

Survey period

The first survey was conducted in July 2011 (2011 Survey), approximately 4 months after the accident. The second survey was conducted 6 months later, in January 2012 (2012 Survey); subsequent surveys were conducted at one-year intervals until January 2016 (2016 Survey). Furthermore, another survey was conducted in January 2022, approximately 11 years after the accident (2022 Survey).

Study areas and participants

Fukushima Prefecture consists of seven administrative districts (Fig. 1). Fukushima City, where the survey was conducted, belongs to the Kenpoku region. Fukushima City is the capital of Fukushima Prefecture and is located approximately 62 km northwest of FDNPP. At the

Table 2
Questionnaires included in the psychological symptom scale for radiation disaster (for children aged 4–12 years) and the results of the exploratory factor analysis.

		F1	F2	F3		
Q1.	She or he becomes silent and does not want to talk.	**0.748**	−0.101	0.032	Depressiveness	α = 0.76
Q2.	She or he has difficulties sustaining interest in new activities in which other children become actively involved.	**0.496**	−0.120	0.222		
Q3.	She or he does not have an appetite for several days.	**0.555**	0.070	0.009		
Q4.	She or he is suppressing her or his emotions.	**0.594**	0.066	0.073		
Q5.	She or he complains about stomachaches, headaches, nausea, sluggishness, etc. when something reminds her or him of a certain event (such as a disaster).	**0.651**	0.159	−0.188		
Q6.	She or he gets startled by sudden noises.	−0.076	**0.822**	0.016	Anxiety and Fear	α = 0.84
Q7.	She or he becomes very fearful at certain triggers.	0.166	**0.622**	−0.018		
Q8.	She or he dislikes being alone.	−0.092	**0.836**	0.052		
Q9.	She or he sometimes clings to adults (refusing to leave her or his guardian).	0.116	**0.623**	0.023		
Q10.	She or he gets irritated and angry or throws tantrums.	−0.026	0.119	**0.600**	Anger and Restlessness	α = 0.65
Q11.	She or he is unable to concentrate on studying or playing.	0.047	−0.013	**0.709**		
		F1	0.579	0.442		
		F2		0.522		

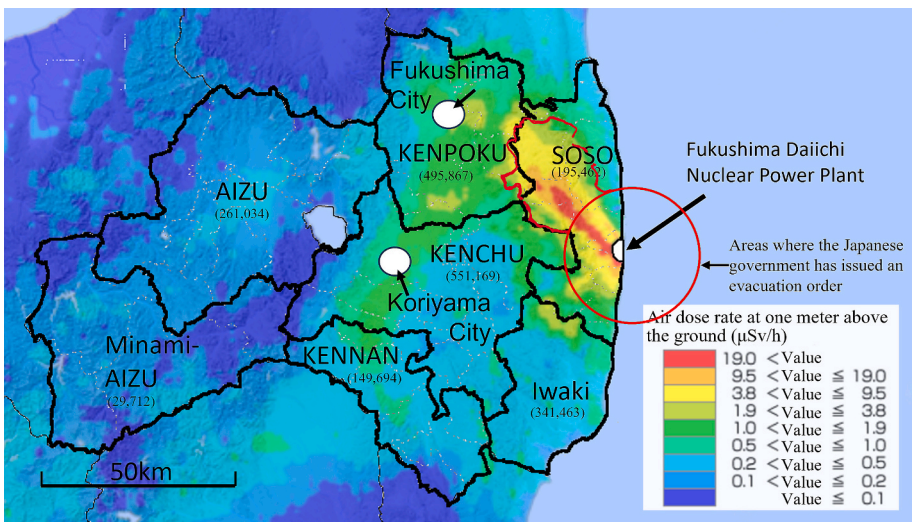

Fig. 1. Location of Fukushima City, Koriyama City, and Fukushima Daiichi nuclear power plant, and the radiation contamination status in Fukushima Prefecture. The aircraft monitoring was conducted by the Ministry of Education, Culture, Sports, Science and Technology, 2011 and the result was partially modified in this figure.

time of the nuclear accident, approximately 290,000 people lived in the area.

The survey was conducted among families with children attending elementary school or kindergarten in Fukushima City. With the cooperation of the Fukushima City Board of Education, which has jurisdiction over local elementary schools and kindergartens, elementary schools and kindergartens were selected each year. Thus, different participants responded to the survey each time. The kindergartens included three types of classes: younger classes with children who turn 4 years old during the year (3 and 4 years old), intermediate classes with children who turn 5 years old (4 and 5 years old), and older classes with children who turn 6 years old (5 and 6 years old). Elementary schools included first graders who turn 7 years old during the school year (6 and 7 years old), second graders who turn 8 years old (7 and 8 years old), third graders who turn 9 years old (8 and 9 years old), fourth graders who turn 10 years old (9 and 10 years old), fifth graders who turn 11 years old (10 and 11 years old), and sixth graders who turn 12 years old (11 and 12 years old). A questionnaire survey was conducted. Questionnaires were distributed through elementary schools and kindergartens, completed at home by the children's mothers, and collected at the elementary school and kindergarten. Attached to the distributed questionnaires were an explanation of the survey, stating that participation was voluntary, and that all data obtained would be managed quantitatively with no risk of identifying individuals, along with a consent form. Written informed consent was obtained from the mothers upon collection of the questionnaire.

In Survey 2013, Survey 2015, and Survey 2022, data were collected in a similar manner in the Kagoshima, Fukui, Hyogo, and Akita prefectures, which were considered less affected by radiation, and Tokyo, which was not significantly affected by radiation[3]. These data were used as the control group, referred to as the Other Prefecture group, for comparison with the situation in Fukushima City. In the surveys conducted in Fukushima City in 2022 and in Other Prefecture group in 2015 and 2022, data collection in kindergartens was not carried out.

Statistical analysis

A one-way analysis of variance with year of survey (seven levels) as the between-subjects factor was used to analyze mothers' self-rated

[3] The approximate distances from the prefectural capitals (or Tokyo Metropolitan Government) to the Fukushima Daiichi Nuclear Power Plant and the number of mother-child pairs used in the analysis are as follows: Kagoshima, 1156 km, 394 pairs; Fukui, 454 km, 411 pairs; Hyogo, 606 km, 921 pairs; Akita, 268 km, 334 pairs; Tokyo, 226 km, 284 pairs.

psychological symptom and radiation protection behavior scores, and children's scores for Depressiveness, Anxiety and Fear, and Anger and Restlessness. To compare the responses of mothers and children in Fukushima City to other prefectures in Survey 2013, Survey 2015, and Survey 2022, analyses were conducted with unpaired *t*-tests for each period.

Furthermore, the unique effects of age, period, and cohort on changes in children's mental reactions were analyzed using APC analysis. Bayesian Age-Period Cohort Modeling and Prediction (BAMP) version 1.3.0.1 (Schmid & Held, 2007), an analysis software specialized for Bayesian APC analysis, was used for the analysis. The top 27 % of the distribution of scores on each psychological scale was used as the reference value (Otomo, 1996), and those with a 73 percentile or higher were considered high responders. The frequency of occurrence of high responders for Depressiveness, Anxiety and Fear, and Anger and Restlessness during the survey period from 2011 to 2016 was analyzed. This study was approved by the Ethics Committee of the relevant university (approval numbers: 2520 and 2020–04).

Results

Although the survey data included a small number of cases in which children's fathers and grandparents responded, only data from mothers' responses were used for statistical analysis[4]. We excluded data on people who lived in the evacuation area at the time of the accident but subsequently evacuated to areas within or outside Fukushima Prefecture, and people who moved to Fukushima Prefecture after the nuclear accident, owing to differences in residential areas during the nuclear accident and the survey. Furthermore, we excluded responses that did not indicate the place of residence at the time of the accident or the child's gender or age. Table 3 shows the number of mother-child pairs used in the final statistical analysis.

Analysis of mothers' psychological reactions

Fig. 2 shows the changes in mothers' psychological symptom scores. Table 4 summarizes the results of the statistical analysis. As the analysis of psychological symptom scores of mothers living in Fukushima City revealed a significant main effect of time ($F(6, 11{,}057) = 73.24, p < .01$), multiple comparisons were performed using the Tukey method. The analysis showed that psychological symptom scores were the highest in 2011 and 2012 and declined over time but increased significantly in 2022 compared to the 2016 results. The scores of psychological symptoms of mothers in Fukushima City were significantly higher than those in other prefectures in 2013, 2015, and 2022 (2013: $t(2818) = 17.34$, $p < .01$, $d = 0.63$; 2015: $t(3097) = 8.51$, $p < .01$, $d = 0.63$; 2022: $t(1168) = 4.06$, $p < .01$, $d = 0.65$).

Fig. 3 shows the changes in the mothers' radiation protection behavior scores. As the analysis of radiation protection behavior scores of mothers in Fukushima City revealed a significant main effect of time ($F(6, 11{,}057) = 448.76, p < .01$), multiple comparisons were performed. The results showed that the radiation protection behavior scores were the highest immediately after the accident and decreased over time until 2016. However, as with the psychological symptom scores, a significant increase was observed from 2016 to 2022. Furthermore, the scores of Fukushima City were significantly higher than those of other prefectures in 2013, 2015, and 2022 (2013: $t(2818) = 44.23$, $p < .01$; 2015: $t(3097) = 24.26$, $p < .01$; 2022: $t(1168) = 4.06$, $p < .01$).

Psychological symptom scores and radiation protective behavior scores in other prefectures in 2015 (pre-COVID-19 pandemic) and 2022 (during the pandemic) were compared using a *t*-test. The results showed that both psychological symptom scores and radiation protection behavior scores were significantly higher in 2022 (psychological symptom scores: $t(1114) = 2.857$, $p < .01$; radiation protective behavior scores: $t(1114) = 10.17$, $p < .01$). However, no difference was found in Fukushima City between 2015 and 2022.

Analysis of children's psychological reactions

Fig. 4 shows the trends in children's Depressiveness scores. As the analysis of Depressiveness scores of children in Fukushima City revealed a significant main effect of time ($F(6, 11{,}057) = 41.65, p < .01$), multiple comparisons were performed. Children's Depressiveness scores were the highest in 2012 rather than in 2011, then declined in 2013, followed by no change until 2015, and then declined again in 2016. No significant differences were found between the 2016 and 2022 results. The scores of Fukushima City were significantly higher than those of other prefectures in 2013, 2015, and 2022 (2013: $t(2818) = 17.19$, $p < .01$; 2015: $t(3097) = 9.79$, $p < .01$; 2022: $t(1168) = 2.35$, $p < .05$).

Fig. 5 shows the trends in children's Anxiety and Fear scores. As the analysis of Anxiety and Fear scores of children in Fukushima City revealed a significant main effect of time course ($F(6, 11{,}057) = 72.24$, $p < .01$), multiple comparisons were performed. Children's Anxiety and Fear responses were the highest in 2011 and decreased year by year. The score was the lowest in 2016, and no significant difference was observed between 2016 and 2022. The scores of Fukushima City were significantly higher than those of other prefectures in 2013, 2015, and 2022 (2013: $t(2818) = 20.51$, $p < .01$; 2015: $t(3097) = 9.66$, $p < .01$; 2022: $t(1168) = 5.55$, $p < .01$).

Fig. 6 shows the trends in children's Anger and Restlessness scores. As the analysis of Anger and Restlessness scores of children in Fukushima City revealed a significant main effect of time course ($F(6, 11{,}057) = 27.83$, $p < .01$), multiple comparisons were performed. Children's Anger and Restlessness responses were high from 2011 to 2012 but declined in 2013 and did not change thereafter. However, the results for 2022 showed a decline and were significantly lower than those for 2016. The scores of Fukushima City were significantly higher than those of other prefectures in 2013, 2015, and 2022 (2013: $t(2818) = 13.49$, $p < .05$; 2015: $t(3097) = 10.59$, $p < .01$; 2022: $t(1168) = 3.46$, $p < .01$).

A t-test comparison of scores in 2015 and 2022 for the Other Prefecture group for children's psychological reactions showed an increase in Depressiveness and Anger and Restlessness scores in 2022 (Depressiveness: $t(1114) = 3.68$, $p < .01$; Anger and Restlessness: $t(1114) = 3.12$, $p < .01$). In contrast, children in Fukushima had lower scores for all mental reactions in 2022 than in 2015 (Depressiveness: $t(3151) = 3.45$, $p < .01$; Anxiety and Fear: $t(3151) = 2.64$, $p < .01$; Anger and Restlessness: $t(3151) = 3.63$, $p < .01$).

Analysis of changes in children's mental reactions using age-period-cohort analysis

We conducted a Bayesian APC analysis using the Bayesian hierarchical cohort model to assess age, period, and cohort effects on annual changes in Depressiveness, Anxiety and Fear, and Anger and Restlessness scores by child age group from 2011 to 2016. Single-factor (A, P, C), two-factor (AP, AC, PC), and three-factor (APC) models were assumed, and the goodness of fit of the models was evaluated by calculating the deviance information criterion (DIC) values. We adopted the APC model after confirming that the DIC values were not significantly different from the minimum. The results of the analysis are shown in Fig. 7.

The period effect was the largest for Depressiveness, whereas the age and cohort effects were relatively small. The period effect increased from 2011 to 2012, peaked in 2012, and continuously declined thereafter. The cohort effect showed a slight but decreasing trend for the cohorts of people born in 2011 and later.

The period effect had the largest influence on Anxiety and Fear, showing a consistent downward trend since 2011. The age effect

[4] Data from cases where someone other than the mother responded were excluded from the analysis, totaling 755 cases for Fukushima City and 133 cases for the other prefectures.

Table 3
Number of mother-child pairs used in the final statistical analysis.

		Gender	Three-year	Four-year	Five-year	Six-year	Seven-year	Eight-year	Nine-year	Ten-year	Eleven-year	Twelve-year	Total
Fukushima-City	2011	boys	15	49	49	26	5	2	7	5	9	2	169
		girls	27	66	43	19	21	23	24	26	28	5	282
	2012	boys	8	39	50	64	98	123	121	128	109	106	846
		girls	11	51	66	72	108	118	136	124	137	109	932
	2013	boys	18	44	48	66	91	85	93	88	84	66	683
		girls	12	26	52	67	108	81	96	103	93	92	730
	2014	boys	12	30	59	73	143	143	133	140	124	128	985
		girls	15	36	46	75	138	173	122	169	161	96	1031
	2015	boys	10	31	47	74	172	199	191	181	179	157	1241
		girls	10	37	56	72	209	189	196	191	214	169	1343
	2016	boys	4	26	49	50	161	145	137	146	114	108	940
		girls	7	17	37	65	162	159	137	149	124	119	976
	2022	boys				3	30	41	41	45	66	43	269
		girls				5	30	35	41	49	59	39	258
Other Prefecture	2013	boys	16	34	40	45	90	103	79	69	66	68	610
		girls	11	28	33	46	97	93	89	94	80	57	628
	2015	boys				1	48	36	42	36	39	31	233
		girls				5	41	38	43	32	36	35	230
	2022	boys				6	39	38	51	63	60	64	321
		girls				8	34	43	45	57	71	64	322

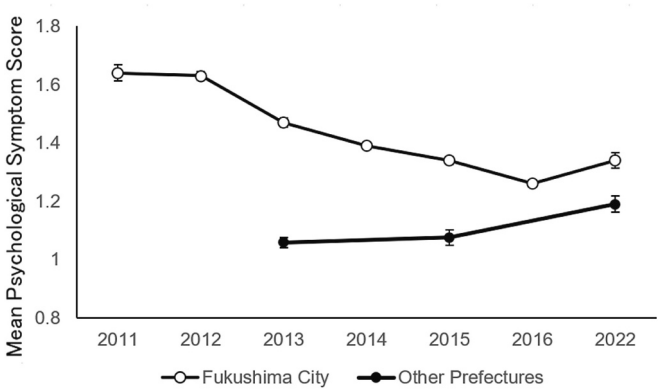

Fig. 2. Trends in psychological symptoms among mothers living in low-contamination areas.

decreased from 3 to 12 years, with a particularly large decrease at 11 and 12 years of age. The cohort effect increased moderately from birth year 2000 to 2006 but leveled off in subsequent cohorts.

Changes in the period effect were the largest for Anger and Restlessness, showing a consistent decline in the effect after 2011. In contrast, changes in the age and cohort effects were negligible.

Discussion

This study analyzed the psychological effects associated with living in low-contamination areas following the FDNPP accident and its long-term effects on 10,158 mother-child pairs living in Fukushima City. The findings suggest that mothers and children who lived in low-dose radiation-contaminated areas after the FDNPP accident showed various psychological reactions related to anxiety, depression, fear, and stress for at least 5 years, although these effects decreased over time. These results were consistent with those of Tsutsui et al. (2020), who conducted a study of 18- and 42-month-old children and their mothers in low-dose radiation-contaminated areas in Fukushima Prefecture immediately after the FDNPP accident. Furthermore, the results were in line with Havenaar et al.'s (1996) report that residents of low-dose areas had mental health problems 6 years after the ChNPP accident. Moreover, these findings were consistent with the results of Sung's longitudinal study, which targeted children who were 3 years old at the time of the accident and their parents living in municipalities in the Nakadori region, including Fukushima City, an area of low-dose radiation contamination (Sung, 2015; Sung et al., 2021).

The findings of this study were not significantly different from those of the MHLS, which investigated children evacuated from designated evacuation zones. However, it is important to note that the MHLS participants were infants and children evacuated from the Evacuation Ordered Zone, and these children experienced a quite complex web of

Table 4
Results of the analysis of variance for changes in psychological effects over 11 years since the accident.

Scores to be analyzed	Results of analysis of psychological effects in Fukushima city		Comparison between Fukushima city and other prefecture group		
	Results of 1way-ANOVA		Results of t-test		
	Main effect	Multiple comparisons(HSD)	2013	2015	2022
Mothers' psychological symptom scores	73.24** (6, 11,057)	2011 = 2012 > 2013 > 2014 > 2015 > 2016 < 2022	t(2818) = 17.34**	t(3097) = 8.51**	t(1168) = 4.06**
Radiation protection behavior scores	448.76** (6, 11,057)	2011 > 2012 > 2013 > 2014 > 2015 > 2016 < 2022	t(2818) = 44.23**	t(3097) = 24.26**	t(1168) = 4.06**
Children's depressive score	41.65** (6, 11,057)	2011⟨2012⟩2013 = 2014 = 2015 > 2016 = 2022	t(2818) = 17.19**	t(3097) = 9.79**	t(1168) = 2.35*
Children's anxiety and fear score	72.24** (6, 11,057)	2011 > 2012 > 2013 > 2014 > 2015 > 2016 = 2022	t(2818) = 20.51**	t(3097) = 9.66**	t(1168) = 5.55**
Children's anger and restlessness score	27.83** (6, 11,057)	2011 = 2012 > 2013 = 2014 = 2015 = 2016 > 2022	t(2818) = 13.49*	t(3097) = 10.59**	t(1168) = 3.46**

Asterisks represent statistical significance (*: $p < .05$, **: $p < .01$).

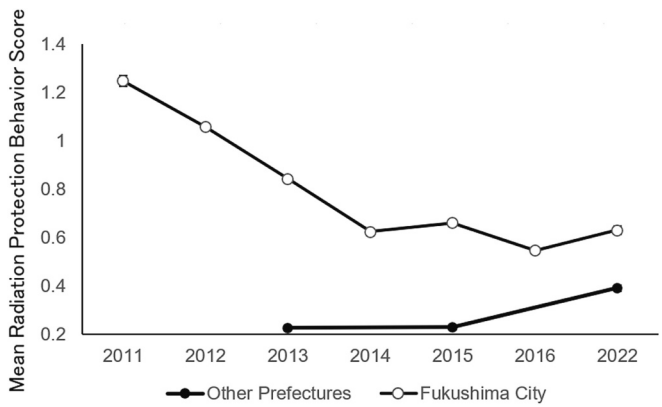

Fig. 3. Trends in radiation protection behavior scores among mothers living in low-contamination areas.

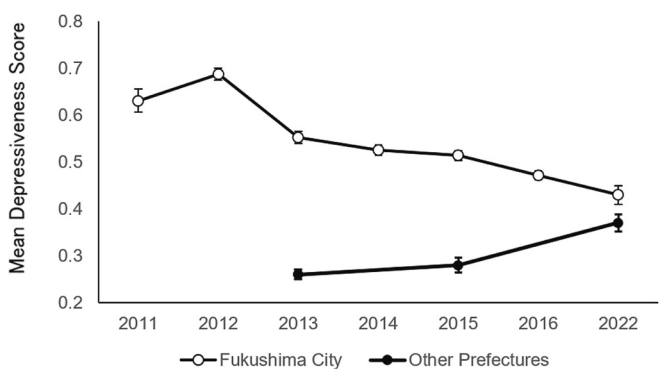

Fig. 4. Trends in depressiveness scores among children living in low-contamination areas.

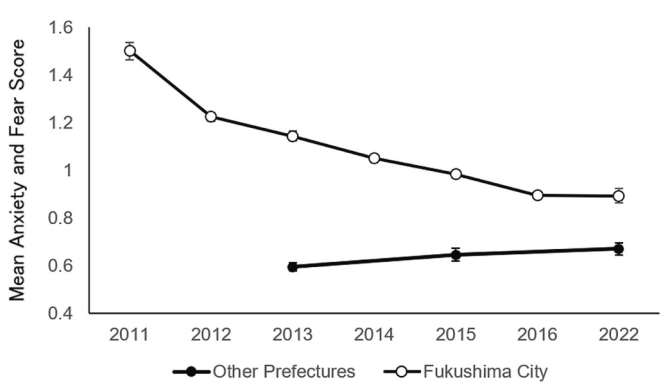

Fig. 5. Trends in anxiety and fear scores among children living in low-contamination areas.

events and long-term difficulties, such as mass and repeated relocation, loss of social and economic stability, and other FDNPP-related stressors. Therefore, directly comparing the results of this study with those of the MHLS may be misleading and should be done with caution. Kamiya et al. (2021) states that 7 years after the accident, the high-risk rate indicated by the SDQ for 4–6 year-olds had improved to a level almost the same as the high-risk rate in a previous study (Matsuishi et al., 2008). Although not reported by parents, the results of the general mental health score using the K6 for adults showed that even 7 years after the accident, the high-risk rate for mood and anxiety disorders remained higher than the national average, especially for women. Kamiya et al. (2021) has been following up on some of the mothers who became pregnant and gave

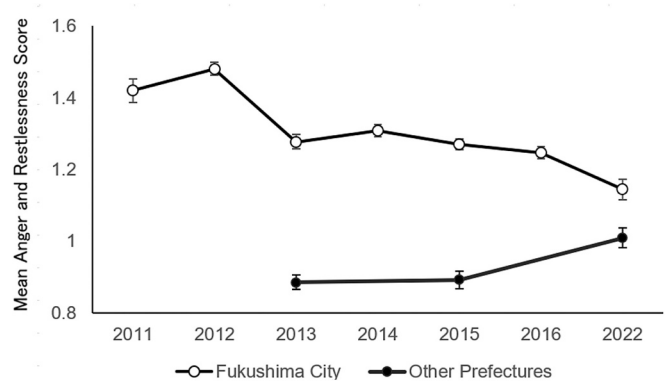

Fig. 6. Trends in Anger and Restlessness scores among children living in low-contamination areas.

birth in Fukushima Prefecture after the accident for four years from the first year. It was revealed that 63 % of the mothers surveyed in the follow-up survey still had concerns about the effects of radiation and their children's health four years after the accident. In contrast, this study was the first to demonstrate that the psychological effects of a nuclear accident persisted in preschool and school children 5 years after the accident.

The findings of this study, which suggest that the mothers studied experienced long-term mental health issues, potentially subclinical, appear to conflict with the explanations and information provided by the Japanese government, the Fukushima Prefecture, and radiation experts, who argue that the level of radiation contamination (dose) in Fukushima City was not significant enough to affect health. Tsutsui et al. (2020) reported that even five years after the accident, mothers living in Nakadori were still taking actions to protect their children from radiation exposure, more so than parents in other prefectures with little to no radiation impact, as also evidenced in Sung (2015). Furthermore, as mentioned earlier, the Prefectural Health Survey on pregnant women and mothers revealed that 63 % still harbored concerns about radiation and their children's health four years after the accident(Kamiya et al., 2021). These findings suggest that the information and risk communication provided by the Japanese government, Fukushima Prefecture, and radiation experts may not have been sufficiently effective in alleviating mothers' concerns about the potential health impacts of radiation exposure on their children. The difficulty of conveying scientifically grounded information to the general public in a way that is both understandable and reassuring has also been observed in the aftermath of the TMI accident (Erikson, 1990) and the ChNPP disaster (Ujiie & Tsutsui, 2022).

This study cannot directly answer the question of why the explanations and information provided by the Japanese government, Fukushima Prefecture, and radiation experts to reassure the public did not sufficiently reassure the mothers. However, It is possible to interpret these findings based on recent insights into risk perception. According to knowledge related to risk perception, dissociation or conflict tend to occur between government disaster management officials or disaster risk experts and the public in their understanding of the situation (Slovic, 1999). The public may tend to consider uncertainty, dread, catastrophic potential, controllability, equity, and risk to future generations in the risk equation (Slovic, 1987), whereas experts tend to view risk as synonymous with probability of harm or expected mortality (Cohen, 1985). Scientists and experts' explanations of safety often do not allay public worries and anxieties (Slovic, 1999) and can sometimes intensify them (Ujiie & Tsutsui, 2022). The sufferers of the ChNPP accident (Havenaar, de Wilde, van den Bout, Drottz-Sjöberg, & van den Brink, 2003; International Atomic Energy Agency, 2006) as well as mothers surveyed by Tsutsui et al. (2020) and in this study, were (or are) living with a real sense of danger. If this is the case, it may be necessary

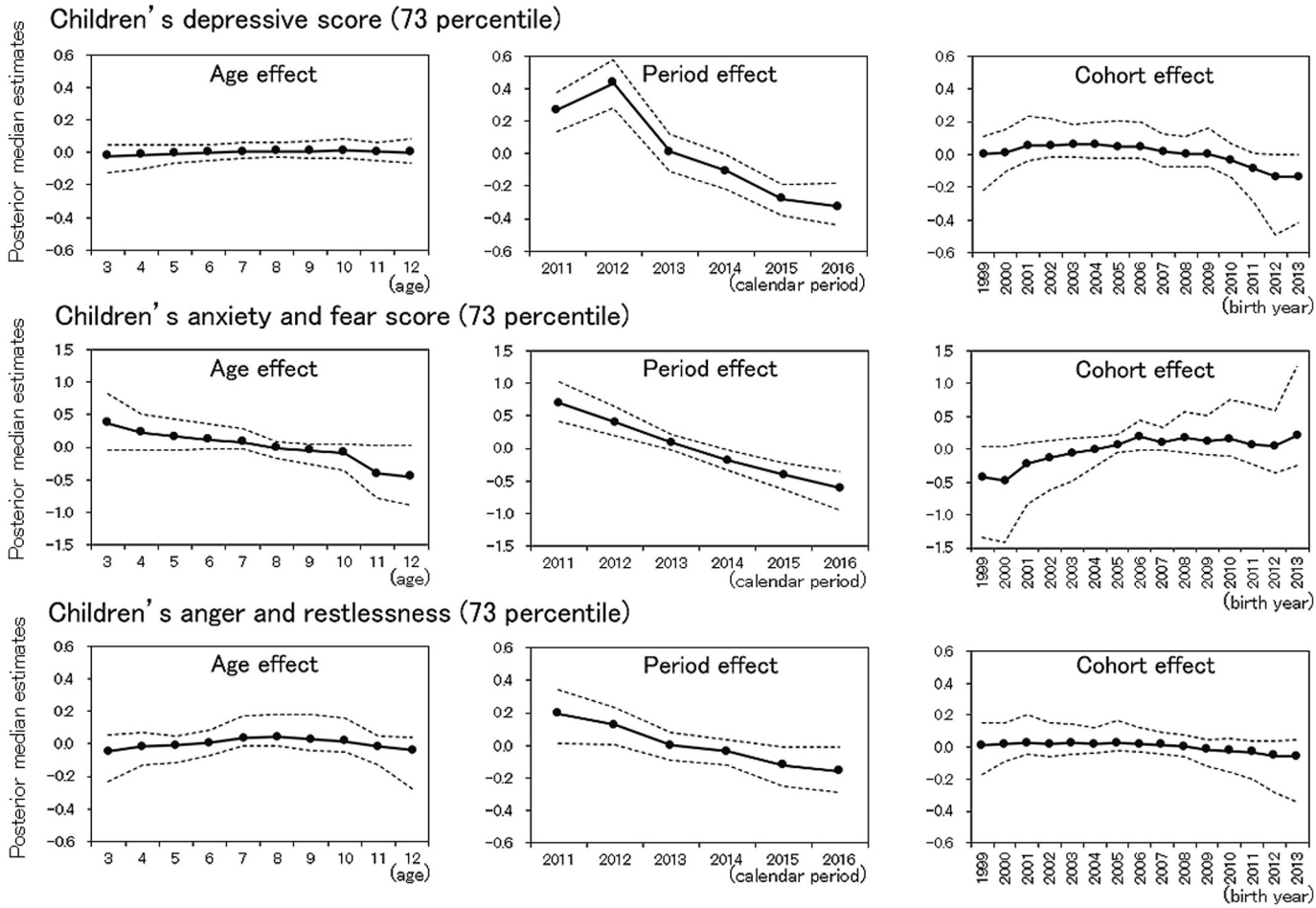

Fig. 7. Estimated values of age, period, and cohort effects on ratio of children's mental reactions (scored the 73rd percentile).
The results of the Bayesian APC analysis are presented as posterior estimates for each time cross-section (age, calendar year, and birth year, respectively) for age, period, and cohort effects, with their medians and 95 % credible intervals (95 %CI) illustrated. Solid lines with circles show estimated posterior median values for age, period, and cohort effects with 95 % CI shown as broken lines indicate the upper or lower limits. A rising trend reflects an increase in risk, whereas a falling trend reflects a decrease in risk. Specifically, a rising trend in posterior estimates means that the trend of the effect increases, affecting ratios in the direction of an increase, while a declining trend in posterior estimates means that the trend of the effect decreases, affecting ratios in the direction of a decrease. The range of posterior estimates for each effect (difference between the maximum and minimum values) indicates the relative magnitude between each effect.

to consider approaches that do not conflict with the public's risk assessment paradigm in order to improve the effectiveness of information dissemination and risk communication by the government and expert organizations, in preparation for the unfortunate event of another severe radiation disaster in the future.

The mental health issues and problems faced by children among evacuees from the nuclear accident, as revealed by the MHLS, are likely due not only to their perception of radiation risk (Miura et al., 2017; Suzuki et al., 2018) but also to the disruption of their living environment and daily activities caused by the evacuation and life in evacuation shelters (Nagai et al., 2017). However, it remains unclear to what extent the disruption of living environments and daily activities can explain the issues observed among the parents and children living in the Nakadori region, a low-dose radiation contamination area and non-evacuation zone, as studied by Tsutsui et al. (2020) and Sung (2015), as well as those issues observed in the parents and children who were the subjects of this study. Indeed, restrictions on outdoor play and concerns over the safety of water and food consumed by children may have caused significant disruptions to daily life. However, it may be reasonable to attribute these issues to parents' anxiety about the possibility that their surroundings, water, and food might be contaminated, as well as their concerns that even minimal radiation exposure could affect their children's health. While the impact of disruptions to the living environment and daily life on the long-term mental health of both children and parents cannot be denied, it is also plausible to consider that underlying these issues is the problem of parents' risk perception.

The 2022 survey was conducted during the COVID-19 pandemic. As of January 2022, Japan was experiencing the sixth wave of the pandemic, although the Japanese government's state of emergency for the pandemic had been lifted. It has been reported that the pandemic had a significant impact on the mental health of mothers and children living in Japan (e.g., Kameda, Ido, Sonoda, Yokoyama, & Hayasaka, 2022; Lin & Yamano, 2023), and it is reasonable to consider that this impact extended to the mothers and children in Fukushima City and the control group from other prefectures who were the subjects of this study. However, a comparison of the results of pre- and post-pandemic surveys showed that the impact of the pandemic was slightly different between Fukushima City and other prefectures. For instance, psychological symptom and radiation protection behavior scores were significantly higher in 2022 than in 2015 among respondents in the Other Prefecture group. Conversely, among respondents in Fukushima City, the difference between 2015 and 2022 was not significant, whereas the difference between 2016 and 2022 was significant. The psychological symptom scores and radiation protection behavior scores of mothers in Fukushima City decreased linearly from 2011 to 2012 to 2016, and the trend continued until 2022. It can be inferred that the psychological symptom

scores and radiation protection behavior scores in 2022 were further influenced by the impact of the pandemic. Similarly, it is inferred that the psychological symptoms related to the nuclear accident in children, like those in their mothers, took about 10 years to decline.

The results of the APC analysis showed that Depressiveness, Anxiety and Fear, Anger and Restlessness among children were predominantly influenced by period effects. This indicates that for children of all ages, the psychological impact was the strongest from immediately after the accident to the following year, and that the impact diminished over time. This result was expected, as Fukushima City was in great turmoil from 2011 to 2012, shortly after the accident. However, considering the cognitive limitations of many children, except those in the upper grades of elementary school, it is unlikely that they showed fear, anxiety, anger, or restlessness due to recognizing the adverse effects of radiation on the human body and the risks. As shown in previous developmental psychological studies on disasters, children were more likely indirectly influenced by their parents, who were confused, anxious, and stressed when faced with radioactive contamination. Children's adaptation during disasters, particularly younger children's, depends on their parents who play an important role in protecting them in such risky situations (Cobham et al., 2016; Gewirtz et al., 2008; Masten & Osofsky, 2010; Silverman & La Greca, 2002). Thus, if parents are psychologically confused, their children are more likely to exhibit a variety of negative psychological reactions. Hence, the period effect in the psychological impact on the children in this study was likely influenced by their parents, who were confused, frightened, anxious, and stressed after the nuclear accident with radioactive contamination.

Disasters disrupt children's usual routines and activities. In Fukushima, children's outdoor activities were restricted from 2011 to 2012, and many children were unable to go outside to play. In kindergartens and schools, schoolyards were decontaminated, topsoil was removed, and monitoring posts were installed to measure radiation levels. We cannot rule out the possibility that the various events that occurred at school, at home, and in the community may have caused the children's psychological disorders.

The cohort effect in psychological symptoms was weaker than the period effect but was observed in Depressiveness and Anxiety and Fear responses. Depressiveness was higher among children born before the accident than those born after it; Anxiety and Fear were relatively low among children born in 2000, but gradually increased year by year for each subsequent birth cohort up to 2006. This suggests that preschool children were the most affected psychologically in 2011, when the accident occurred, whereas the impact on children who were already enrolled in school may have been relatively small. Previous developmental psychological research on disasters reported that the impact of disasters varied depending on the age of the child when it occurred (Franks, 2011b; Masten et al., 2015; Narayan & Masten, 2012; Peek et al., 2018; Silverman & La Greca, 2002). However, unique cohort effects have not been examined in previous studies on nuclear disasters. This study was the first to show that the psychological effects, specifically "Depressiveness" and "Anxiety and Fear" responses to a nuclear disaster, differed depending on birth cohort and the age of the children when it occurred.

The finding that preschool children had the strongest relative psychological impact in 2011 and that the impact on children who were already in school was relatively small may be related to the extent of children's dependence on their parents. Young children are highly dependent on their parents, and children try to interpret whether a situation is safe through their parents' reactions (Gewirtz et al., 2008). If parents perceive the situation as dangerous and express anxiety, fear, or nervousness about contamination, young children are more likely to perceive the situation as dangerous and feel insecure. In contrast, as children reach school age, they have access to resources outside of their parents and families, such as teachers, peers, and school counselors, and can utilize their own cognitive and non-cognitive resources (Masten & Osofsky, 2010). Thus, they may be more resilient to the effects of environmental stresses and parental psychological disorders and inadequate protections.

In developmental psychology, it is widely accepted that mothers should be the target of psychological support to mitigate the negative impacts of disasters on child development (e.g., Cobham et al., 2016; Gewirtz et al., 2008; Halevi et al., 2017). In response to the Great East Japan Earthquake and the nuclear accident, Fukushima Prefecture and Fukushima Medical University have conducted surveys and provided support, particularly targeting high-risk residents (Kamiya et al., 2021). The Prefectural Health Survey includes a section focused on pregnant women who became pregnant and gave birth in Fukushima after the accident, where various surveys and support measures have been implemented (Goto, 2018; Kamiya et al., 2021). Additionally, Fukushima Prefecture established the Fukushima Mental Health Care Center to address mental health issues among residents, offering individual support particularly to those with high mental health risks (Hiruta, 2014; Maeda & Watanabe, 2018). According to Maeda and Watanabe (2018), the initial assumption was to consider all residents of Fukushima as disaster victims. However, in reality, the number of potential support recipients was overwhelmingly large, while the resources required for support were limited, necessitating the selection of support recipients based on the severity of their disaster exposure. It is well known that parents of young children are vulnerable in the face of disasters (Bromet & Havenaar, 2002). However, it can be said that the disaster exposure and mental health risks of the participants in this study, as well as those in the studies by Tsutsui et al. (2020) and Sung (2015), were not as high as those of evacuees. Nevertheless, it is plausible to imagine that the potential need for support among parents living in low-dose radiation-contaminated areas was not insignificant. In Japan, major disasters have continued to occur even after the Great East Japan Earthquake, and unfortunately, there is a possibility that nuclear disasters could occur again in the future. In preparation for such situations, it may be necessary to recognize the potential support needs of mothers with young children, and to establish mechanisms that enable support even within limited resources, ensuring that psychological support is available when needed.

This study demonstrated that children who were not in school during the accident suffered from the strongest psychological impacts. These children are now entering adolescence. Various negative childhood experiences resulting from the nuclear accident may have had a particular impact on the children's subsequent developmental process. Individuals who were children at the time of the FDNPP accident were restricted from outdoor activities in the early years of schooling. They will continue to be concerned about the probabilistic health effects of radiation exposure for a long time. Furthermore, as more children leave their parents' homes to pursue higher education or other opportunities outside the prefecture, they may face bullying and stigmatization, similar to what evacuees from Fukushima experienced in their host communities (Oe et al., 2019; Sawano, Ozaki, Hori, & Tsubokura, 2019; Tsubokura, 2020). The continued effects of these factors on the personality formation and identity development of children raised in Fukushima City should be monitored.

Study limitations

The study measured children's psychological reactions using their mothers' responses. According to Masten and Osofsky (2010), mothers are more likely to overestimate their children's symptoms compared to objective observations and may report symptoms that differ from those reported by the children themselves. Furthermore, mothers with poor mental health could have skewed the responses regarding children's psychological reactions; mothers who perceive themselves or their children as being affected by the accident may overestimate the psychological impact on their children. The importance of measuring experiences and reactions to disasters directly from the children themselves has been increasingly recognized (Peek et al., 2018). This

study included children who were of a sufficient age to self-report. However, one of the objectives of this study was to determine the psychological effects of living in low-contamination areas on children based on their age, which required using the same methods for all surveys. Furthermore, we were practically constrained by the extreme difficulty of collecting data from children regarding issues of radiation.

Moreover, this study used a repeated cross-sectional survey rather than a longitudinal survey. This hindered the estimation of trends in the mental health of individual mothers and children living in Fukushima based on the results of this study. However, as this study aimed to identify changes in the psychological conditions of the mothers and children in Fukushima over time following the nuclear accident, we believe the repeated cross-sectional survey and APC analysis adequately served that purpose.

Authors' statement

I, Yuji Tsutsui, the corresponding author of the work certify that all authors have participated sufficiently in the conception and design of the work and were involved in the acquisition of data. All of them have also participated in the analysis and interpretation of the data, as well as drafting the work. All authors also revised the work and contributed to the final approval of the version for publication. Each author also agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Ethics approval

This study was approved by the Ethics Committee of Fukushima University (approval numbers: 2520 and 2020-04).

CRediT authorship contribution statement

Yuji Tsutsui: Writing – review & editing, Writing – original draft, Visualization, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tatsuo Ujiie:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Conceptualization. **Rieko Takaya:** Project administration, Investigation, Formal analysis, Data curation. **Hiroyuki Uchida:** Writing – original draft, Supervision, Software, Resources, Data curation. **Youichi Odagiri:** Writing – original draft, Supervision, Software, Resources, Data curation. **Misako Tominaga:** Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Madoka Takahara:** Investigation, Formal analysis, Data curation.

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Data availability

Data will be made available on request.

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